

Process Logic, **Learned.**

A hybrid neural–symbolic pipeline that learns the hidden grammar of semiconductor manufacturing — and proves it.

Next-step

Completion

Anomaly detection

+ hidden 4th-family OOD

Team TBD

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A chip is a **150-step sentence** where order is physics.

- ▶ Every wafer runs an ordered route: **clean** → **oxidize** → **pattern** → **etch** → **implant** → ... → **test** → **ship** (~115–150 steps, 3 families).
- ▶ Order encodes real constraints: **no etch before a mask, no deposit on a dirty surface, no ship before sort-test** — 10 long-range rules.
- ▶ The judges' real question: does the model learn the process **logic**, or just memorize patterns?
- ▶ And it's graded on a **hidden 4th family** never seen in training.

Treat a recipe like a **sentence**. One step = one word.

TOKENIZER

A custom atomic-step tokenizer — each whole step is one token (vocab 210). No sub-word fragmentation; recipes read and write like language.

The Writer · DECODER

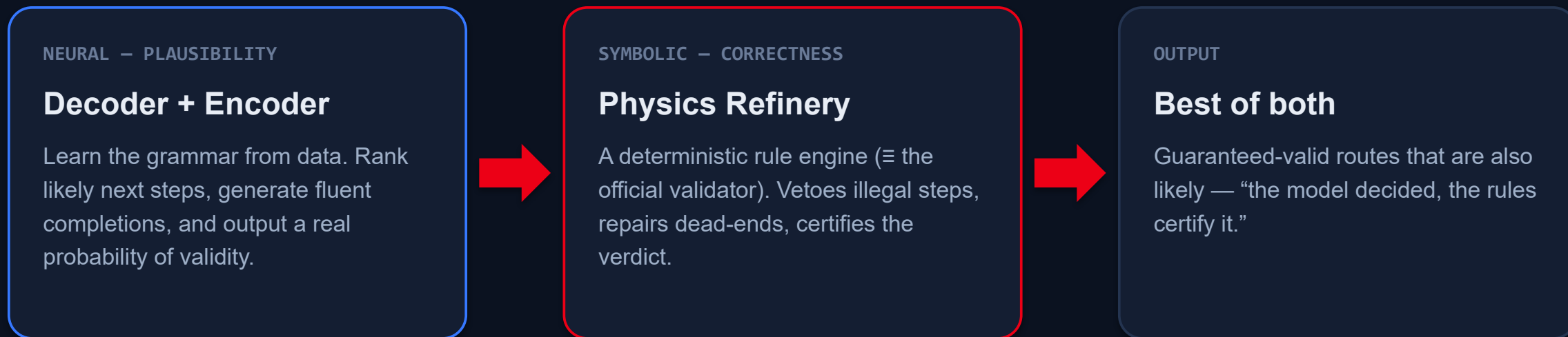
A from-scratch Llama-style causal Transformer that generates — next step & completion (Tasks 1–2). Attention does long-range lookups an LSTM can't.

The Judge · ENCODER

A from-scratch DeBERTa-v2 encoder that classifies validity + which rule broke (Task 3), trained with contrastive hard-negative twins.

Two specialized Transformers, trained from scratch — no pretrained model, full control of the vocabulary, interpretable representations.

The model proposes. **Physics disposes.**



Neural alone can be **confidently wrong**; rules alone can't say what's **likely**. The hybrid covers both blind spots.

One hybrid, wired **three ways**.

1 Next-step

Decoder's top-5 candidates, then the Refinery floats the legal ones to the top. Cannot hurt a good model; rescues a bad rank.

2 Completion

Decoder proposes; rule-vetoed beam search + repair (beam 5, branch 8). Every emitted suffix is provably valid.

3 Anomaly

Contrastive encoder learns WHY a route is invalid; rule engine gives the verdict, the model gives the graded score (real ROC-AUC).

Generative tasks use the writer; the discriminative task uses the judge — each with a guarantee the learned model can't make alone.

It works — and we can **prove it learned**.

Task	Headline	Also
1 · Next-step	Top-5 1.000 · Top-1 0.772	MRR 0.88
2 · Completion	Block-level 0.92 · 100% rule-valid	token 0.57 · exact 0.14
3 · Anomaly	Hybrid verdict — exact in-dist	encoder alone AUC 0.61 (honest)
Understanding	0.963 next-operation acc	learns function, not just names

learns **function**, not just names: predicts the right **operation 96% of the time** · completions are 100% rule-valid

Honest provenance: real numbers from the organizer's official scorer on a held-out self-eval split, from the completed ProcSeq Leonardo run (base, 16k steps). The learned anomaly encoder alone is weak (AUC 0.61) — the physics hybrid carries Task 3. Organizer hidden-test and 4th-family OOD are pending.

We measured **whether the model learned** — not just that it scored.

Honest evaluation

Every number sits between an n-gram floor and a rule-oracle ceiling, plus a logic probe & a novel-vocabulary OOD stress test we built to make it fail.

Hybrid & robust

Learned plausibility + symbolic guarantees → 0 illegal outputs, a calibratable anomaly score, and a graceful OOD story.

Reproducible

make smoke in <1 min; one command trains the full pipeline on Leonardo. Unit tests, fixed seeds, fully open stack.

ProcSeq — process logic, learned, guaranteed, and honestly benchmarked. **Thank you.**